NESA NATIONAL EXAMINATION AND SCHOOL INSPECTION AUTHORITY

SWD - Back-End Development and Database

T2-138

Monday, 14 July 2025

8:30 AM to 11:30 AM

Names:.....

Index number :.....

TSS NATIONAL EXAMINATIONS, LEVEL 5, 2024-2025

QUESTIONS and ANSWERS BOOKLET

TRADE (s): SOFTWARE DEVELOPMENT (SWD)

SUBJECT: BACK-END DEVELOPMENT AND DATABASE

DURATION: 3 Hours

Read carefully the instructions on page 3

TOTAL MARKS...../100

FOR EXAMINER'S USE ONLY

QUESTIONS	1	2	3	4	5	6	7	8	9	10	Total
Marks											

QUESTIONS	11	12	13	14	15	16	17	18	19	20	Total
Marks											

QUESTIONS	21	22	23	24	25	26	27	28	29	30	Total
Marks											

TSS NATIONAL EXAMINATIONS, LEVEL 5, 2024-2025

TRADE: SOFTWARE DEVELOPMENT (SWD)

SUBJECT: BACK-END DEVELOPMENT AND DATABASE

DURATION: 3 HOURS

INSTRUCTIONS TO CONDIDATE:

1. This Exam paper is composed of **three sections (A, B and C).** Follow the instructions given below and answer the indicated questions for a total of **100 marks.**

   Section **A:** Fourteen (14) questions which are **Compulsory** **55 marks**

   Section **B:** Among the five (5) questions, attempt any three (3) **30 marks**

   Section **C:** Among the two (2) questions, attempt anyone (1) **15 marks**

2. Fill in your actual **names** and **Index number** on the provided space(Cover).
3. Do not remove or tear any page or part of this Questions and Answers Booklet.
4. Answer in the language in which the Examination is set.
5. In section **A**, answer questions in the provided space. In case of multiple-choice questions, just **circle** the letter corresponding to the correct option.
6. In section **B** and **C**, answer questions in the provided space after section C questions.
7. **Allowed materials:**
   - Blue or Black pen.
   - Geometrical Instruments
   - Silent non-programmable calculators

SECTION A: Attempt all questions **(55 marks)**

---

01. Read the sub-questions below and circle the correct answers among the options given: **(4Marks)**

i. What is the primary purpose of an Entity-Relationship Diagram (ERD)?

a) To represent data movement in a system

b) To show relationships between entities in a database

c) To define decision-making rules

d) To illustrate user interactions with a system

ii. Why flowcharts are commonly used in system design?

a) To manage computer hardware

b) To visually represent an algorithm or process

c) To define system security policies

d) To create web applications

02. Using the table below match the backend system components with their correct descriptions: **(3Marks)**

Component	**Description**	**Answer**
1. API	A. Handles business logic between the user and database.	**1 = ...**
2. Database server	B. Stores and manage data.	**2 = ...**
3. Application server	C. Provides the interface for communication between software.	**3 = ...**

03. Circle the letter corresponding to the answer to the sub-questions among the options given below. **(4Marks)**

i. Which of the following is a benefit of using data encryption in securing RESTful APIs?

a) Increase API response time.

b) Ensure data confidentiality during transmission.

c) Reduces the need for authentication.

d) Prevents the use of third-party libraries

ii. Which node package is commonly used to manage environment variables securely in backend applications.

a) Express
b) Passport
c) dotenv
d) Winston

04. Circle the letter corresponding to the answer to the sub-questions among the options given below. **(4Marks)**

i. What is the main purpose of an Admin Panel in a CMS?

a) To view website analytics
b) To edit and manage content
c) To manage user comment
d) To design the website layout

ii. Which of the following is commonly used to store content in a CMS?

a) JSON files
b) MySQL databases
c) HTML files
d) PHP scripts

05. Read the questions below and circle the correct answer to each of the following sub-questions. **(3Marks)**

i. What is the primary purpose of Flutter's widget Tree in UI design?

a. To store database records
b. To manage application States
c. To define the hierarchical structure of UI components
d. To handle user authentication

ii. Which of the following is **NOT** a core widget in Flutter?

a. Text
b. Image
c. Button
d. Firewall

iii. Which state management technique is used in Flutter for managing UI states efficiently?

a. Redux

b. SQL Query

c. REST API

d. JavaScript

06. Circle the letter corresponding to correct answer to each of the following sub-questions: **(4Marks)**

i. Which of the following is required for publishing an App on the Apple App Store?

a. Developer Profile

b. App Store Connect account

c. Mobile Device with iOS

d. Email account

ii. Which of the following tools is primarily used for submitting Android apps to Google Play?

a. Xcode

b. Android Studio

c. Google Play Console

d. Eclipse

07. Read the statements below and answer with **True (T)** if the statement is correct, or **False (F)** if the statement is incorrect. **(4Marks)**

a. MongoDB allows both embedding documents within other documents and using references between documents to establish relationships……

b. When designing MongoDB collections, the only proper way to establish relationships is through embedded documents……

c. In MongoDB, you should embed documents when the data is frequently accessed together and has a strong belonging relationship…….

d. In NoSQL database, data is always stored in a fixed schema like in relational database……

08. Circle the letter corresponding to correct answer to each of the following sub-questions: **(4Marks)**

i. Which of the following is a characteristic of a normalized table?

a) It must have only one column

b) It should only have primary keys

c) Each column must have atomic values

d) Each record must be stored in a separate database

ii. Which of the following best describes a one-to-many relationship in a database?

a) One record in a Table A is related to only one record in a Table B.

b) Many records in Table A are related to one record in Table B.

c) One record in Table A can be related to multiple records in Table B.

d) Each record in Table A must match exactly one record in Table B.

09. Circle the correct answer to each of the following questions: **(4Marks)**

i. Why is it important to use HTTP status codes like 200, 201, and 404 in a RESTful API?

a. To specify the error message

b. To identify the database schema

c. To inform the client about the outcome of the request

d. To limit the number of API calls

ii. What does the following code snippet do in a Node.js Express app?

```
app.use((err, req, res, next) => {
      res.status(500).send('Something
went wrong!');
        });
```

a. It logs the errors to a file

b. It handles all errors and sends a generic error message

c. It validates input data

d. It handles successful HTTP requests

10. Using the table below match the **testing types** in column A with their corresponding **descriptions** in column B: **(4Marks)**

Column A: Testing Types	**Answer**	**Column B: Descriptions**
1. Unit Testing	1 = ...	a)Testing the interaction between different backend components.
2. Integration Testing	2 = ...	b) Testing how well the backend handles large numbers of concurrent users.
3. API Testing	3 = ...	c) Testing individual functions or methods in isolation.
4. Performance Testing	4 = ...	d)Testing the functionality, reliability, and security of backend APIs.

11. Circle the letter corresponding to correct answer to each of the following sub-questions: **(5Marks)**

i. A database administrator wants to give a user read-only access to a table named Employees. Which SQL command should be used?

a. GRANT INSERT ON Employees TO user;
b. GRANT UPDATE ON Employees TO user;
c. GRANT SELECT ON Employees TO user;
d. GRANT DELETE ON Employees TO user;

ii.You are tasked with securing sensitive customer information stored in a database, which approach would you apply to ensure that even if the database is accessed, the data is unreadable?

a. Use of indexing
b. Data replication
c. Data encryption
d. Data normalization

iii. A developer needs to prevent SQL injection attacks in a web application that accesses the database, which of the following should they apply in the code?

a. Dynamic SQL queries
b. Plain-text credentials
c. Parameterized queries or prepared statements
d. Allowing all user inputs

iv. You need to ensure that only authorized users can access the production database during work hours, what should you implement?

a. Backup scheduling
b. Role-based access control and time-based restrictions
c. Schema normalization
d. Use of audit triggers

v. A company requires that any changes made to critical financial data be tracked. What should be implemented to meet this requirement?

a. Full-text indexing
b. Database mirroring
c. Auditing and logging
d. Foreign key constraints

12. Read the questions below and circle the correct option among the given to each sub-question. **(4Marks)**

i. Which of the following is the correct way to insert data into a database using PDO in PHP?

a) $conn->query("INSERT INTO users VALUES ('John', 'john@example.com')");
b) $conn->prepare("INSERT INTO users (name, email) VALUES (?, ?)"); $conn->execute(['John', 'john@example.com'])
c) mysqli_query($conn, "INSERT INTO users VALUES ('John', 'john@example.com')");
d) pdo_insert($conn, "INSERT INTO users (name, email) VALUES (?, ?)", ['John', 'john@example.com']);

ii. What is the purpose of the htmlspecialchars() function in PHP?

a) To escape HTML characters and prevent Cross-Site Scripting (XSS) attacks.

b) To sanitize SQL queries.

c) To generate HTML code for web pages.

d) To encode data before inserting into a database.

13. Circle the best answer to each sub-question below: **(4Marks)**

i. How do you generate a URL for a named route in Laravel?

a. url('profile')

b. route('profile')

c. generate('profile')

d. Route::url('profile')

ii. Which command is used to install Laravel using Composer among the following?

a. composer create-project --prefer-dist laravel/laravel project-name

b. composer install laravel

c. composer start-laravel project-name

d. laravel new project-name/

14. Referring to knowledge of Mobile App Development, match each feature from **Column A** with the appropriate use case in **Column B**. **(4Marks)**

Column A: Flutter feature	**Answer**	**Column B: Use Case**
1. setState()	1 =	A. Managing global app state
2.Navigator.push()	2 =	B. Handling complex UI updates locally
3. Provider	3 =	C. Navigating between screens
4. Container	4 =	D. Styling and layout of widgets

Section B: Attempt any three (3) questions **(30 marks)**

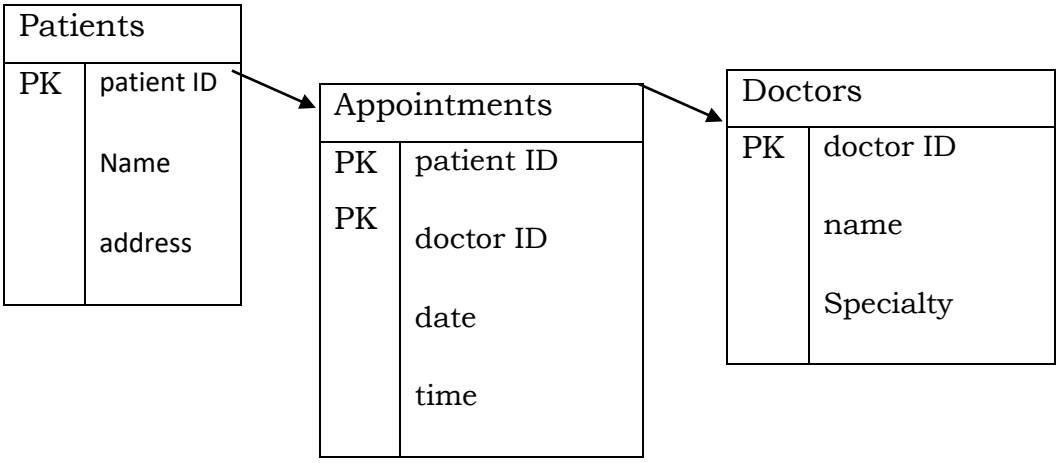
---

15. Kaneza is a system analyst at a healthcare technology company. He has been assigned to document the workflow of a new patient record management system that integrates multiple departments, including reception, labs and billing. His team needs to clearly understand how data moves across these departments and how various processes interact. To effectively communicate the system design to both developers and stakeholders, Kaneza must decide whether to use a Data Flow Diagram (DFD) or a Flowchart. **(10marks)**

"With the help of a table, analyze the difference between Data Flow Diagrams (DFDs) and Flowcharts in representing complex system interactions."

16. A hospital wants a database to track patients, doctors and appointments. Analyze the following case study and answer questions that follow: **(10marks)**

- Each patient has a patientID, Name and Address.
- Each Doctor has a DoctorID, Name and Specialty.
- Each appointment connects one patient to one Doctor and records the Date and Time.



Questions

i. Which type of relationship exists between Patient and Appointments?

ii. Name the primary key in the Patients table

iii. Identify the entities, attributes, and relationships needed for conceptual schema.

iv. Explain how to transform the conceptual design into a logical database design by listing tables and key constraints.

17. Analyze how different http request methods (GET, POST, PUT, DELETE) would be applied in a mobile application that manages user accounts. **(10marks)**

"For each method, explain the specific scenario where it would be appropriate and highlight potential risks or challenges when handling server responses for each method."

18. Bigwi leads a high school project team developing a Node.js backend application for managing student attendance and grading. As the team approaches the deployment phase, they must choose the most suitable method to host and run their application. Bigwi is tasked with evaluating different deployment approaches to help the team decide between setting up everything manually on a server or using Docker for containerized deployment. **(10marks)**

i. Evaluate Manual Deployment and Docker-based Deployment for a NodeJS backend application by briefly explaining them, and discussing their advantages and disadvantages.

ii. Based on your evaluation, recommend which deployment method would be better for a high school project team and justify your choice.

19. Muneza is developing a PHP-based web application for a local business to manage inventory and sales data. As he begins interacting with database, she must choose between using MySQLi and PDO for connecting her PHP application to the database. To decide, Muneza needs to understand how each method works and weigh their pros and cons. **(10marks)**

"Evaluate the use of **MySqli** and **PDO** for connecting PHP applications to a database by explaining each method, and discussing their advantages and disadvantages."

Section C: Attempt only one (1) question **(15 marks)**

---

20. A company has hired you as a junior database developer to build for them a simple internship management system. **(15marks)**

The database should store the following:

- **Students** (student_id, full_name, university, major, phone_number).
- **Companies** (company_id, company_name, industry)
- **Internship** (intership_id, student_id, company_id, start_date, end_date, internship_status).

i. Create a database **InternshipDB** and three tables above with appropriate data types and constraints.

ii. Insert at least one records into each table.

iii. Write a query to find students currently doing internships in the software industry.

iv. Grant select permission on the students table to a user **intern_manager**.

21. A school has hired you to build a simple Bookstore Inventory database using MongoDB for their library. The system should manage: **(15marks)**

- **Books** (title, author, genre, quantity, price)
- **Members** (name,phone,phone_number, email)

***Tasks*:**

i. Create a database **LibraryDB** and two collections: **Books** and **Members**.

ii. Insert two documents into each collection.

iii. Write a query to find all books with quantity less than 5.

iv. Create an index on the **author** field in the **Books** collection.

v. Drop the Members collection.

END

RESERVED SPACE FOR SECTION B AND C ANSWERS

Do not write in this margin

Do not write in this margin

Do not write in this margin

Do not write in this margin

Do not write in this margin

Do not write in this margin

Do not write in this margin

Do not write in this margin

Do not write in this margin